

Matrix discrepancy and optimization

The sharp universal Spencer discrepancy constant

Difficulty: extreme **Importance:** interesting to the community

Status: Open

Provenance: source-stated extremal-value problem.

Rating rationale: Determining the best constant across all sign matrices and all orders is a longstanding extremal discrepancy barrier; it sharpens a central matrix-balancing theorem.

For each positive integer n , define

$$d_n = \frac{1}{\sqrt{n}} \max_{A \in \{-1,1\}^{n \times n}} \min_{x \in \{-1,1\}^n} \|Ax\|_\infty.$$

Determine the exact value of

$$C_* = \sup_{n \geq 1} d_n.$$

Thus C_* is the smallest real constant for which every square sign matrix A admits a column signing x with $\|Ax\|_\infty \leq C_* \sqrt{n}$.

The question concerns the sharp worst-case error in simultaneous signed sums. In matrix form it asks how far a square sign matrix can keep all sign vectors from the coordinate cube of radius $C\sqrt{n}$.

The source also poses $\limsup_{n \rightarrow \infty} d_n > 1$ as a related asymptotic conjecture. It is recorded here as context rather than counted as another entry.

References

1. A. S. Bandeira, A. Kireeva, A. Maillard, and A. Rödder, *Randomstrasse101: Open Problems of 2024*, arXiv:2504.20539 (2025), entry 6, Open Problem 11, Conjecture 12, and the subsequent “Updates” paragraph. [Paper](#).
2. A. S. Bandeira, *Did just a couple of deviations suffice all along? (problems 10–14)*, Randomstrasse101 (2024), the original statement of Open Problem 11. [Author’s research post](#).
3. J. Spencer, *Ten Lectures on the Probabilistic Method*, second edition, SIAM (1994), Chapter 10, “Six Standard Deviations Suffice,” pp. 75–80, the classical universal-constant upper bound. [Chapter](#).

Status check — 2026-09-10

checked the current arXiv record, still 2504.20539v1 (2025-04-29), the source’s update, and searches “Spencer sharp constant discrepancy” and “Spencer limsup discrepancy 2026”. The update refutes the stronger odd Sylvester–Hadamard claim in Conjecture 13; that claim is excluded. No resolution of Open Problem 11 or Conjecture 12 was found. Recent matrix Spencer results concern a different matrix-valued discrepancy problem.

Audit update (2026-09-10): Rechecked Open Problem 11 and the source’s update, then searched for exact Spencer constants. The refuted odd Sylvester–Hadamard claim is different. The all-order sharp supremum warrants extreme rather than hard. This is a bounded literature check, not a proof that no solution exists.